

Stat 331/531: Midterm Exam Short Answer

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```
library(tidyverse)
library(knitr)
```

```
wage_data <- read_csv("us-minimum-wage-data.csv")
party_data <- read_csv("us-party-data.csv")
```

Question 1.

```
min_wage_dat <- wage_data |>
  mutate(
    year = as.integer(str_sub(Effective.Date, 1, 4)),
  ) |>
  select(year,
         State,
         Footnote,
         Federal.Minimum.Wage.2020.Dollars,
         Effective.Minimum.Wage.2020.Dollars,
         State.Minimum.Wage.2020.Dollars)
```

Question 2.

```
min_wage_dat |>
  filter(Footnote == "(b)") |>
  nrow() |>
  kable()
```

—
x
—
80
—

```
#Google Search: r how to get number of rows from df  
#https://www.digitalocean.com/community/tutorials/get-number-of-rows-and-columns-in-r
```

Question 3.

```
min_wage_dat |>  
  filter(year >= 2010, year <= 2020) |>  
  group_by(year, State) |>  
  summarise(ratio = mean(State.Minimum.Wage.2020.Dollars/Federal.Minimum.Wage.2020.Dollars),  
            State = State,  
            StateMinWage = State.Minimum.Wage.2020.Dollars,  
            FedMinWage = Federal.Minimum.Wage.2020.Dollars) |>  
  group_by(year) |>  
  summarise(max(ratio)) |>  
  kable()
```

year	max(ratio)
2010	1.179070
2011	1.195444
2012	1.246022
2013	1.267081
2014	1.310606
2015	1.448799
2016	1.586428
2017	1.586928
2018	1.827309
2019	1.929155
2020	1.931034

```
# For Some reason I cant get it to show the required variables of State and the wages :(
```

Question 4.

```
min_wage_dat |>  
  filter(str_detect(State,
```

```

    "^M")) |>
group_by(State) |>
summarise(mean(State.Minimum.Wage.2020.Dollars)) |>
rename("MeanMinimumWage" = "mean(State.Minimum.Wage.2020.Dollars)") |>
arrange(desc(MeanMinimumWage)) |>
kable()

```

State	MeanMinimumWage
Massachusetts	9.236415
Maine	8.633962
Maryland	8.259623
Michigan	8.128868
Minnesota	6.952830
Montana	6.203396
Missouri	4.460377
Mississippi	0.000000

```

#r rename columns
#https://dplyr.tidyverse.org/reference/rename.html

```

Question 5.

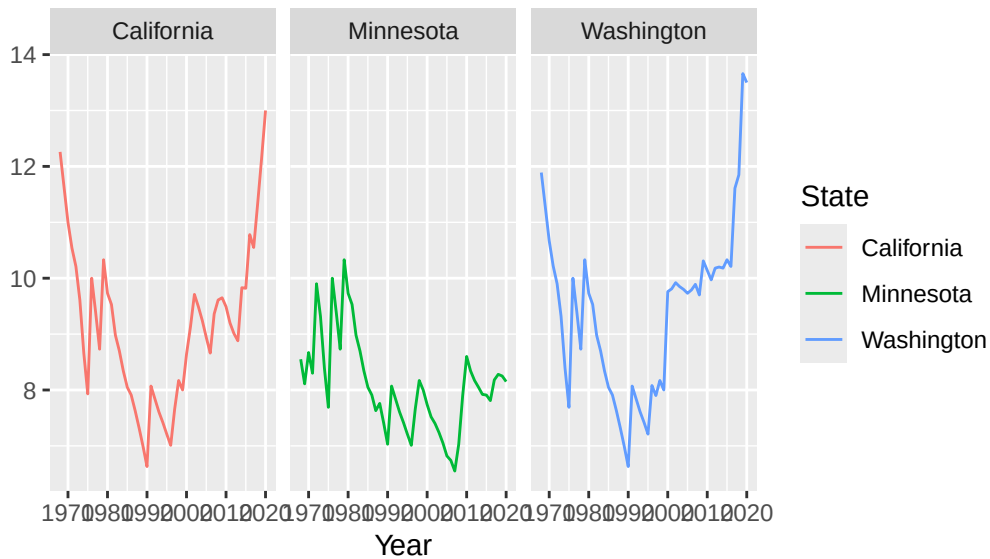
```

min_wage_dat |>
  filter(State == "California" | State == "Washington" | State == "Minnesota") |>
  ggplot(mapping = aes(x = year,
                        y = Effective.Minimum.Wage.2020.Dollars,
                        color = State)) +
  geom_line() +
  facet_wrap(~ State) +
  labs(
    title = "Minumum Wage Comparison (2020 dollars)",
    x = "Year",
    y = "",
    subtitle = "Effective Minumum Wage"
  )

```

Minumum Wage Comparison (2020 dollars)

Effective Mininum Wage



Question 6.

```
min_wage_dat |>
  select(State,
         year,
         State.Minimum.Wage.2020.Dollars,
         Effective.Minimum.Wage.2020.Dollars,
         Federal.Minimum.Wage.2020.Dollars) |>
  rename("state" = "State.Minimum.Wage.2020.Dollars") |>
  rename("federal" = "Federal.Minimum.Wage.2020.Dollars") |>
  rename("effective" = "Effective.Minimum.Wage.2020.Dollars") |>
  pivot_longer(
    cols = c(state,
              effective,
              federal),
    names_to = "Wage_Type",
    values_to = "Money"
  ) |>
  slice_head(n = 9) |>
  kable()
```

State	year	Wage_Type	Money
Alabama	1968	state	0.00
Alabama	1968	effective	8.55
Alabama	1968	federal	8.55
Alaska	1968	state	15.61
Alaska	1968	effective	15.61
Alaska	1968	federal	8.55
Arizona	1968	state	3.48
Arizona	1968	effective	8.55
Arizona	1968	federal	8.55

Question 7.

```
party_dat <- party_data |>
  mutate(party = str_to_title(party)) |>
  mutate(support_strength = factor(case_when(
    perc_votes > 0.55 ~ "Strong",
    perc_votes > 0.50 ~ "Moderate",
    .default = "Weak"
  )))
```

used ?factor in terminal

The code below will confirm that you created **support_strength** correctly. Please leave it in your submission!

```
party_dat |>
  pull(support_strength) |>
  levels()
```

```
[1] "Moderate" "Strong"   "Weak"
```

Question 8.

```
renamed_dat <- min_wage_dat |>
  rename("state" = "State")

renamed_dat |>
  left_join(party_dat, by = c("state", "year")) |>
  select(year, state, everything()) |>
```

```
slice_tail(n = 5) |>
kable()
```

year	state	Foot- note	Federal.Min- imum.Wage.2020.Dol- lars	Effective.Min- imum.Wage.2020.Dol- lars	State.Min- imum.Wage.2020.Dol- lars	party	perc_vot- er	test_strength
2020	Vir- ginia	[c]	7.25	7.25	7.25	Demo- crat	0.541095	Moder- ate
2020	Wash- ing- ton	NA	7.25	13.50	13.50	Demo- crat	0.579703	Strong
2020	West Vir- ginia	NA	7.25	8.75	8.75	Re- pub- lican	0.686315	Strong
2020	Wis- con- sin	NA	7.25	7.25	7.25	Demo- crat	0.494495	Weak
2020	Wyoming	NA	7.25	7.25	5.15	Re- pub- lican	0.694995	Strong